

For dynamic programming, individuals would not make these decisions directly. Instead, a utility function would be determined, which provides a link to the satisfaction the individual derives from different amounts of spending and legacy. This utility function would build in a degree of spending flexibility about how willing the individual is to spend more today with the trade-off that this could require less spending in the future. Given the mathematics of this trade-off, the remainder of the strategy will follow from the dynamic programming methodology about how much to spend each year and how to adjust asset allocation each year as circumstances evolve. The results are meaningful to the extent that the mathematical function expressing the trade-offs between spending certainty and the average level of spending—as well as between spending and legacy—adequately reflects the preferences of the individual using the model.

Irlam and Tomlinson build a case study for a conservative individual with \$500,000 of financial assets, \$20,000 of Social Security benefits, and no particular desire to leave a legacy. They compare outcomes for the solution offered by the dynamic programming methodology to the outcomes for various decision rule and actuarial methods and asset allocation strategies. For spending, they consider constant inflation-adjusted spending strategies, a fixed-percentage strategy, and a strategy similar to the RMD rule spending. For asset allocation, they consider fixed allocation strategies, age in bonds, age minus ten in bonds, and target-date funds.

Among the other strategies, they find that the closest match to their optimal dynamic programming solution is the RMD-styled strategy with a fixed 90 percent stock allocation. Next is a 6.8 percent fixed-percentage strategy with a 90 percent stock allocation. In third is a constant inflation-adjusted spending strategy with a 60 percent stock allocation. All the other asset allocation strategies result in worse outcomes because they lack sufficient aggressiveness. It is interesting that the variable spending strategies are accompanied by higher stock allocations, the reason being that varying withdrawals provide a safety value for stock volatility. With constant inflation-adjusted spending, a lower stock allocation to reduce volatility is the only avenue for alleviating sequence risk.